Robert Laughlin

DATA ENGINEER

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Open to remote work and relocation to Iowa

PROFILE

Entry-level data engineer who built a public Google Cloud pipeline: Terraform-provisioned Cloud Storage and BigQuery, a Dockerized Python ingestion job with retries and change detection (403,264 sale records), and a dbt star schema with 205 tests. Strong SQL and Python, with Harvard's CS50 SQL certificate.

CORE SKILLS

Languages: SQL, Python, Bash; also C++, TypeScript, Go
Data: BigQuery, dbt, Cloud Storage, MySQL, SQLite, indexes and query plans, dimensional modeling (star schemas)
Infrastructure: Terraform, Docker, GCP, Linux, Git and GitHub
Python: pandas, NumPy, requests, google-cloud-bigquery, google-cloud-storage, uv

SELECTED PROJECTS

Polk County Residential Sales Pipeline | **2026**
Python, SQL, BigQuery, Cloud Storage, Terraform, Docker, dbt, uv | GitHub
- Provisioned a versioned Cloud Storage bucket, a raw BigQuery dataset, and one dataset per dbt layer with Terraform, and generated the dbt profile and sources from Terraform outputs.
- Wrote a Python ingestion job (uv, Docker) that downloads the Assessor's yearly sale files with retries, compares MD5 hashes against Cloud Storage to upload only new or changed files, and loads 403,264 sale records (1990-2026) into an 80-column raw BigQuery table.
- Built 38 dbt models ending in a star schema: two sale-level fact tables, parcel, building (versioned per sale), date, and code dimensions, and bridge tables for multi-parcel sales.
- Added 205 dbt tests, enforced contracts on 32 models, and source freshness checks. Husky hooks run ruff, sqlfluff, yamllint, and ShellCheck on every commit; a dry-run script estimates each model's BigQuery scan.

Analytical SQL | **2026**
SQL, SQLite, MySQL, Harvard CS50 SQL
- Completed Harvard's CS50 Introduction to Databases with SQL (certificate), with a final project that loads a 2,000-sale sample of the pipeline's star schema into SQLite.
- Practiced window functions, conditional aggregation, as-of joins, and join-cardinality checks; fixed a fan-out bug by aggregating to one row per sale before joining.

NeoTUI | **2026**
Shell, Lua, tmux, zsh, lf, Neovim | GitHub | Site
- Built a local-first terminal environment (tmux, zsh, lf, Neovim) with isolated configuration, version-aware installation, and pre-commit ShellCheck and Lua checks; used daily.

Neural Network from Scratch | **2021**
Python, NumPy, MNIST | GitHub
- Implemented backpropagation and stochastic gradient descent with NumPy matrix operations; about 95-96% accuracy on the 10K-image MNIST test set.

EDUCATION & CERTIFICATIONS

City University of Seattle | **Coursework toward B.S. in Data Science | 2019-2021**
100 quarter credits. Coursework: foundational statistics, statistical computing, calculus, discrete mathematics, database technologies, and computer science.

Certifications:
- CS50's Introduction to Databases with SQL, Harvard University (2026)
- Introduction to Statistics, Stanford University via Coursera (2025)
- Probability & Statistics for Machine Learning & Data Science, DeepLearning.AI via Coursera (2025)
- Programming with Google Go Specialization, 3 courses, UC Irvine via Coursera (2025)
- Deep Learning Specialization, 5 courses, DeepLearning.AI via Coursera (2024)
- Calculus: MITx 18.01.1x and 18.02.1x, verified, edX (2020-2021); Mathematics for Machine Learning: Multivariate Calculus, Imperial College London (2021)

Self-study: MIT OCW 6.041 Probabilistic Systems Analysis (2026), with worked problem sets.
Problem solving: Advent of Code 2021 and 2022 (all puzzles; 2022 solutions in C++ on GitHub); 170+ LeetCode problems.

Current focus: sale-price analysis on the Polk County marts, and an ingestion log table in BigQuery.